

Technical Projects & Systems Portfolio

A concise briefing of software, analytics, ERP, artificial-intelligence and systems-design work, including live links, project status, architecture and technical stack.

PROFILE	FOCUS	BASE
Analyst Executive Support	Data, ERP, AI-enabled workflows	Accra, Ghana Remote-ready

Portfolio at a glance

Project	Status	Primary domain	Public link
AGL OPS Command Center	LIVE	ERP / operational analytics	https://agl.software
Adinkra Intelligence Systems	ACTIVE / PROTOTYPE	Analytics / AI interface	https://blkkstarhub.com/adinkra-front
BLKK Autos	MVP / VALIDATION	Cross-platform commerce	https://blkkstarhub.com/autos-app
Blkk Star Hub	PLATFORM BUILD	Digital trade / multi-tenant OS	https://blkkstarhub.com
MIRACLE	LIVE	Gamified life operating system	https://miracleapp.site
Unified Family Educational ERP	ACTIVE DEVELOPMENT	AI learning / multi-tenant ERP	github.com/Adwoa-Acheampong/Family-Edu-System
Building AGL	CONCEPT	AR diagnostics	Concept / no public demo
Portfolio Website	LIVE	Personal portfolio / web	https://aacheampong.com

Status discipline: Implemented technologies are stated as implemented. Where the wider Blkk architecture is still planned or being built, the stack is explicitly labelled designed/planned rather than presented as production-deployed.

1. AGL OPS Command Center

LIVE**Role:** Business Analyst / Digital Integrations Advisor / Systems Lead**Link:** <https://agl.software>

Product brief

A live, browser-based ERP and staff operations environment for Automobiles Ghana Ltd. It replaced fragmented paper and spreadsheet workflows with a searchable operating system for workshop activity, vehicle inventory, sales, expenses, staff attendance, HR records and management reporting.

What I designed / delivered

- Business-process and requirements architecture across workshop, sales, inventory, expense, attendance and staff-record workflows.
- Kanban-based work-order tracking, QR-code workspace authentication and role-based dashboards for different operational users.
- Structured data-capture and reporting practices that give management a consistent view of operational and commercial performance.
- Staff training and adoption workflows, connecting system design to actual day-to-day use rather than treating the platform as a standalone software build.

Technical / systems stack

Application type	Web-based ERP / staff hub with operational dashboards
Core patterns	Role-based access, QR authentication, Kanban workflow management, searchable structured records
Data domains	Workshop job cards, vehicles, sales, inventory, expenses, attendance, HR and management KPIs
Analytics layer	Operational dashboards, performance reporting, data validation and decision-support views
Framework note	The exact production frontend/backend framework is not documented in the portfolio materials used for this briefing, so it is not overstated here.

Why it matters

AGL OPS is the strongest evidence of systems work moving beyond prototype thinking into a real operational environment. It demonstrates requirements analysis, workflow redesign, data structuring,

user adoption, executive reporting and business-impact measurement in one project.

2. Adinkra Intelligence Systems (AIS)

**ACTIVE /
PROTOTYPE****Role:** Lead Developer / Engineer / Product Architect**Link:** <https://blkkstarhub.com/adinkra-front>

Product brief

The analytics and intelligence frontend for Blkk Star Hub. Adinkra is designed to sit above tenant ERP environments and convert authorised operational data into shared analytics, reporting and artificial-intelligence-assisted decision support. A tenant such as AGL can therefore be viewed from both its ERP operations side and its analytics side.

Implemented technical stack

Frontend	React 19, Vite
UI / styling	Tailwind CSS v4, Framer Motion
Routing	React Router
Data visualisation	Recharts
Operational tooling	QR workflows; spreadsheet parsing/import/export via SheetJS (XLSX)
Architecture	Multi-tenant analytics interface connected conceptually to Blkk Star Hub's shared backend and central artificial-intelligence layer

Key engineering highlights

- Built a data-rich interface for multi-vertical operational metrics and tenant drill-downs.
- Used Recharts for business visualisation and SheetJS for spreadsheet interoperability, supporting the reality that many SMEs still operate partly through Excel-based workflows.
- Designed the interface as the decision layer above ERP activity rather than as an isolated dashboard product.
- Provides the visual foundation for role-specific intelligence for merchants, managers, accountants, financial institutions and investors as the backend matures.

3. BLKK Autos

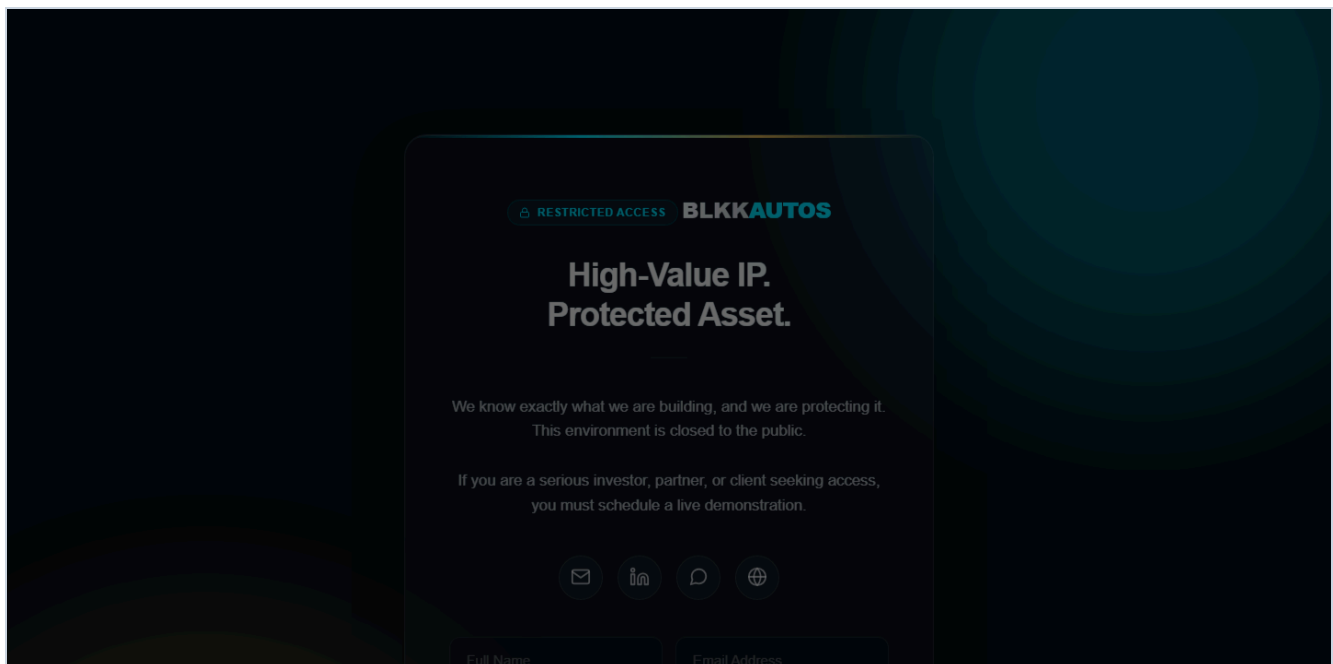
MVP /
VALIDATION

Role: Frontend / Mobile Engineer / Product Architect

Link: <https://blkkstarhub.com/autos-app>

Product brief

A universal automotive-commerce application for web, iOS and Android and the first sector-specific validation wedge for Blkk Star Hub. It structures vehicle listings, dealer operations, buyer discovery and transaction workflows while providing a path for verified commercial activity to feed the wider data and analytics architecture.



Public access status: The BLKK Autos page is intentionally locked as a protected MVP asset. The landing page states that the environment is closed to the public and directs serious investors, partners or clients to request a live demonstration.

Implemented technical stack

Application	React Native 0.86
Cross-platform runtime	Expo 57
Routing	Expo Router
Styling	Tailwind CSS

Motion / interaction	React Native Reanimated and gesture-based interaction
Deployment pattern	Single codebase targeting iOS, Android and web; static web bundle / micro-frontend integration within the broader Blkk Star Hub web experience

Key engineering highlights

- One application architecture serving mobile and web rather than maintaining separate platform codebases.
- Routing and deep-linking designed for both native navigation and web deployment.
- Micro-frontend positioning allows the automotive vertical to operate independently while remaining part of the larger Blkk Star Hub ecosystem.
- Product design is built around the automotive trust problem: structured listings, verification, dealer workflows and transaction history rather than classifieds alone.

4. Blkk Star Hub

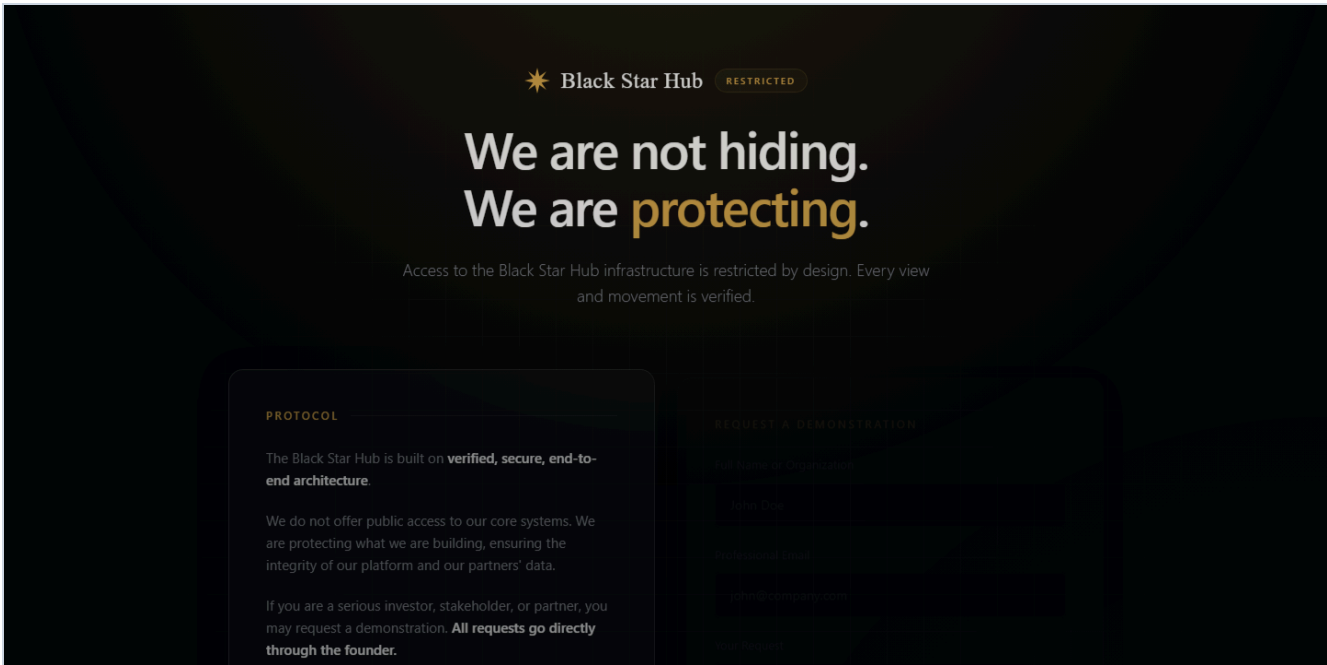
PLATFORM BUILD

Role: Founder / Strategist / Systems Architect

Link: <https://blkkstarhub.com>

Product brief

A modular, API-first digital trade and business operating system designed to help African MSMEs move from fragmented, locally trusted commerce into structured, verifiable and increasingly financeable operations. The system combines sector-specific commerce frontends with a shared institutional spine for identity, operations, data, analytics, verification, payments, logistics and artificial intelligence.



Public access status: The Black Star Hub landing page is restricted by design to protect core infrastructure, platform integrity and partner data. Demonstration requests are routed through founder-led review rather than open public browsing.

Current executable components

- BLKK Autos: cross-platform automotive-commerce MVP.
- Adinkra Intelligence Systems: shared analytics and intelligence interface.
- AGL OPS: real-world operational ERP pattern informing the tenant operating-system layer.

Designed / planned platform architecture

Web / mobile	React / Next.js web architecture; React Native / PWA patterns for mobile-first access
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Primary data	PostgreSQL with JSONB for flexible cross-vertical catalogue and operating records
Search	Typesense or Elasticsearch for unified cross-vertical search
Performance	Redis for caching/session management; RabbitMQ for asynchronous events
Identity / API	JWT authentication, API Gateway and modular service boundaries
Distributed workflows	Microservice-oriented services with Saga-pattern orchestration for cross-service transactions
Payments / commerce	Designed payment orchestration, escrow-state logic, MoMo/card rails and split-payout workflows
Intelligence	Adinkra as the central analytics / artificial-intelligence layer consuming permissioned tenant and transaction data

Sector modules

The broader architecture includes Autos, Merchant, Farmers and Estate verticals. These remain at different maturity levels and should be read as sector-specific expressions of one shared data, trust and operating-system architecture rather than as four unrelated applications.

5. MIRACLE

LIVE**Role:** Product Designer / Builder**Link:** <https://miracleapp.site>

Product brief

A mobile-first life operating system built around faith, focus, stewardship and follow-through. MIRACLE turns personal disciplines such as prayer, study, goals and routines into a more engaging, game-like experience through progress, rewards and reflective interaction.

Product / interaction stack

Experience	Mobile-responsive web application
Interaction model	Gamification, progress tracking, rewards, routines and reflective content
Design direction	Personal dashboard and game-like UX intended to make consistent action more emotionally engaging
Framework note	The exact production framework, backend and database are not documented in the project materials available for this briefing, so no unsupported framework claim is made.

What the project demonstrates

- Product thinking outside enterprise software, including motivation, habit formation and emotionally resonant interaction design.
- Ability to move from personal problem framing into a coherent digital product concept and usable live experience.
- Experimentation with richer interfaces and game mechanics that complements the more operational and analytics-heavy systems in the portfolio.

6. Unified Family Educational ERP

ACTIVE
DEVELOPMENT

Role: Systems Designer /
Builder

Link: github.com/Adwoa-Acheampong/Family-Edu-System

Product brief

A multi-tenant, AI-powered family learning ecosystem that gives different family members age-appropriate learning interfaces while using shared infrastructure underneath. The design combines a React/PWA dashboard, Google Workspace services and an AI orchestration layer for content generation, assignments, knowledge retrieval, grading and progress tracking.

Current repository stack

Frontend	React 19, Vite 6, TypeScript
UI	Tailwind CSS 4, Motion, Lucide React
State / data	TanStack React Query, React Router DOM 7
Visualisation	Recharts 3
AI SDK	Google GenAI
Edge / server	Cloudflare Workers / Wrangler, itty-router, Express, esbuild
Build / tooling	TypeScript, Bun lockfile / npm-compatible scripts, Vite, esbuild

Documented architecture / integrations

- Google Drive for storage, Google Classroom for LMS/assignment workflows and NotebookLM as a knowledge-retrieval layer.
- Local AI Engine Room concept with FastAPI, Ollama/local models and OpenRouter model routing.
- Multi-tenant authentication designed around user isolation and Google OAuth.
- Offline-ready PWA direction and personalised UI personas for different ages and learning goals.

7. Building AGL

CONCEPT**Role:** Concept Designer / Systems Architect

Product brief

An augmented-reality mechanic diagnostic concept designed to bring structured vehicle fault data into the technician's field of view. The idea combines goggle-mounted camera input with fault-code information to guide diagnosis and repair in real time, while creating structured service evidence that can feed operational records.

Concept stack

Interface	Augmented-reality / wearable display concept
Inputs	Goggle-mounted camera feed plus vehicle fault-code / diagnostic data
Intelligence	AI-assisted interpretation and guided diagnostic workflows (planned)
Systems integration	Designed to connect diagnostic activity with AGL operational / vehicle-service records

Supporting digital assets & research

Portfolio website

<https://aacheampong.com> - Personal portfolio deployed through GitHub Pages with a custom domain. The deployed build is a React single-page web application.

GitHub

<https://github.com/Adwoa-Acheampong> - Public profile and repositories, including the Unified Family Educational ERP source repository.

Research & systems architecture

Formalising Africa's Informal Economy: Blkk Autos as the Entry Engine for an AI-Enabled Pan-African Trade Operating System (2026). Independent working paper and systems-research project examining the legal, commercial, trust, data-governance and technical architecture required for accountable digital trade infrastructure. Available on request.

Portfolio through-line: Across these projects, the recurring pattern is the conversion of fragmented human activity into structured systems: capturing better data, designing usable workflows, surfacing decision-ready information and progressively introducing artificial intelligence where it can reduce friction without replacing necessary human judgement.